CV of Atousa Kalantari

Google Scholar: https://scholar.google.com/citations?user=78ERVYAAAAAJ&hl=en
GitHub: https://github.com/Atousa-Kalantari

Education

- Continuous Master's Track Student (B.Sc. + M.Sc.) 2018-2026

  Institute for Advanced Studies in Basic Sciences (IASBS), Zanjan, Iran

  Overall GPA: 17.40/20

  Relevant Coursework: *Fluid Physics, Numerical Analysis, Differential Equations, C++ Programming, Computational Physics, Machine Learning, Electrodynamics, Astrophysics, Computational Astrophysics, Galactic Dynamics, Stellar Dynamics, Gravitation, Cosmology*

- **Master's Thesis:** Microlensify: A Transformer-Based Machine Learning Classifier for Microlensing Events Trained on TESS Light Curves

  Advisors: Prof. Hosein Haghi, Dr. Somayeh Khakpash

  Thesis Grade: 20/20

Technical Skills

- **Programming Languages:** Python (Advanced), C++ (Intermediate), C# (Basic), Julia (Basic), MATLAB (Basic)
- **Machine Learning:** Transformer-Based Models, Physics-Informed Variational Autoencoders (VAEs), Self-Organizing Maps (SOMs), K-Means Clustering, Random Forest, t-SNE
- **Tools & Software:** LaTeX (Intermediate), Git (Intermediate)

Developed Packages

- **Microlensify:** Open-source Python package for detecting and classifying microlensing signals, extracting event durations from the latent space. Developed as part of my M.Sc. thesis. [GitHub link]

Publications

- **Kalantari, A.**, Khakpash, S., Sajadian, S., Haghi, H., Fortino, W. F., and Di Stefano, R. “Microlensify: A Transformer-Based Machine Learning Classifier for Microlensing Events Trained on TESS Light Curves.” *The Astrophysical Journal Supplement Series* (under revision).
- Sajadian, S., **Kalantari, A.**, Fatheddin, H., and Khakpash, S. “Simulating Gravitational Microlensing Events by TESS: Predictions on Statistics and Properties.” *The Astronomical Journal*, 169(1), 34 (2025).
- Abedini, F., Gozaliasl, G., Hasani Zonoozi, A., **Kalantari, A.**, Korpi-Lagg, M., Ilbert, O., et al. “COSMOS-Web: Estimating Physical Parameters of Galaxies Using Self-Organizing Maps.” *Monthly Notices of the Royal Astronomical Society*, 548(1), stag337 (2026).
- Asadi, V., Hasani Zonoozi, A., Haghi, H., Abedini, F., **Kalantari, A.**, Jafariyazani, M., and Chartab, N. “Leveraging Machine Learning for Accurate and Fast Stellar Mass Estimation of Galaxies.” *The Astrophysical Journal*, 989(1), 65 (2025).
- Gozaliasl, G., Yang, L., Kartaltepe, J. S., et al., ..., **Kalantari, A.**, et al. “Brightest Group Galaxies in COSMOS-Web: Evolution of the Size-Mass Relation Since z = 3.7.” *Astronomy & Astrophysics*, 703, A129 (2025).

Contributed Talks and Posters

- Remote Talk (in English): Cosmology from Home, October 2026.
- Remote Talk (in English): 28th International Microlensing Conference, February 2026.
- Remote Talk (in English): Rubin observatory microlensing group meeting, January, 2026.
- Remote Talk (in English): 2025 COSMOS Meeting, Marseille, France.
- Poster: 18th National Conference on Astronomy and Astrophysics of Iran, February 2025, Shiraz University, Shiraz, Iran.
- Poster and Remote Short Talk (in English): 27th International Microlensing Conference, January 2025.
- Talk: 26th Meeting on Research in Astronomy, May 2024, IASBS, Zanjan, Iran.
- Poster and Remote Short Talk (in English): 26th International Microlensing Conference, February 2024.
- Poster and Short Talk: 17th National Conference on Astronomy and Astrophysics of Iran, February 2024, Shahid Beheshti University, Tehran, Iran.

Teaching Experience

Teaching Assistant, Introduction to Python Programming for Astronomy Spring 2025

Instructor: Dr. Somayeh Khakpash

Mentoring Experience

Co-Mentor: Two students in a project on microlensing
Mentor: Somayeh Khakpash

Organizational Experience

LOC Member in 25th Meeting on Research in Astronomy, IASBS, Zanjan, Iran, 14–16 June 2023.

Workshops, Schools, and Conferences Attended

- 2-Hour Workshop on "The Astrolabe": Prof. Jan P. Hogendijk, Wilfred de Graaf, Tom Reijngoudt, and Pouyan Rezvani (Utrecht University, The Netherlands), IASBS, Zanjan, Iran, April 2018. (In person)
- "21st Meeting on Research in Astronomy": IASBS, Zanjan, Iran, 9–10 May 2018. (In person)
- 60-Hour Workshop on "Programming in C++": Tehran Institute of Technology, Tehran, Iran, July–September 2018. (In person)
- "23rd Meeting on Research in Astronomy": 28–29 May 2020. (Virtual)
- Introduction to Machine Learning Workshop: IPM, Tehran, Iran, 10 March and 21 April 2022. (Virtual)
- "3rd INO Workshop on Exoplanets and Their Identification": 13–14 January 2022. (Virtual)
- "24th Meeting on Research in Astronomy": IASBS, Zanjan, Iran, 15–17 June 2022. (In person)
- School of Statistical Analysis of Cosmic Fields: September 2022. (Virtual)
- "25th Meeting on Research in Astronomy": IASBS, Zanjan, Iran, 14–16 June 2023. (In person)
- "Summer School on Particle Physics Workshop": ICTP, Trieste, Italy, 19–30 June 2023. (In person)
- "2023 Michigan Cosmology Summer School": 5–9 June 2023. (Virtual)

- "The Transient and Variable Universe Conference": 20–22 June 2023. (Virtual)
- "Cosmology from Home": 29 June–14 July 2023. (Virtual)
- "17th National Conference on Astronomy and Astrophysics of Iran": Shahid Beheshti University, Tehran, Iran, February 2024. (In person)
- "26th International Microlensing Conference": February 2024. (Virtual)
- "26th Meeting on Research in Astronomy": IASBS, Zanjan, Iran, May 2024. (In person)
- Summer School in Statistics for Astronomers: Penn State, June 2024. (Virtual)
- ZTF Summer School 2024: Machine Learning for Time-Domain Astronomy: July 2024. (Virtual)
- 2024 Sagan Summer Workshop: August 2024. (Virtual)
- PHYSTAT–Statistics Meets ML: Imperial College, September 2024. (Virtual)
- "27th International Microlensing Conference": January 2025. (Virtual)
- "18th National Conference on Astronomy and Astrophysics of Iran":Shiraz University, Shiraz, Iran, February 2025. (In person)
- "2025 COSMOS Meeting": May 2025, Marseille, France. (Virtual)
- ZTF Summer School 2025: Astronomy Data Science in the Rubin Era: July 2025. (Virtual)
- "Cosmology from Home": 22 September–2 October 2026. (Virtual)

References

- **Dr. Somayeh Khakpash:**
  Department of Astronomy, University of Virginia, USA
  Email: somayeh.khakpash@gmail.com

- **Prof. Hosein Haghi**:
  Department of Physics, Institute for Advanced Studies in Basic Sciences, Iran
  Email: haghi@iasbs.ac.ir